Rutuja Dugaje

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Summary

AI/ML Developer with a strong specialization in **Generative AI, Agentic AI** and **Large Language Models (LLMs)**. Proven ability to build intelligent, real-time AI solutions, including **chatbots**, **voice assistants** and **video insight systems**. Proficient in **LangChain**, **OpenAI APIs**, **Gemini**, **FastAPI, Flask** and developing scalable **RAG pipelines**. Experienced in **containerization with Docker**, **cloud deployment on AWS**, and implementing **CI/CD pipelines** for reliable and automated workflows. Seeking to leverage expertise as an **AI Engineer**, **NLP Developer** or Backend **Python Developer** to deliver intelligent and real-time AI solutions.

Professional Experience

AI/ML Developer Intern *May 2025 - Present*

Application Square Infotech, Nashik

- Designed and developed dynamic **chatbot** and **voicebot** applications using **LLMs**, **LangChain**, and **RAG** techniques for natural context-aware interactions on web and voice platforms.
- Integrated real-time **speech recognition** and **TTS** using **gTTS**, **Google Speech API, Whisper** and **pyttsx3**.
- Built scalable **FastAPI** and **Flask backends** with modular chatbot endpoints supporting low-latency **ML model inference** and **retrieval-based generation**.
- Containerized applications with **Docker** and deployed ML services on **AWS** for seamless scalability and integration.
- Created custom **AI pipelines combining embeddings**, **retrievers, memory chains**, and **RAG workflows** to deliver accurate and context-sensitive responses.

Projects

Medical Chatbot – MediCare | *Python, Flask, LangChain, Docker, AWS, Gemini API, Github, ReactJS, Tailwind CSS*

- Developed a medical chatbot using **LangChain**, **embeddings**, and **retrieval-based generation** for context-aware health conversations.
- Built scalable **Flask backend APIs** with modular endpoints supporting real-time ML model inference.
- Containerized the application with **Docker** and **deployed on AWS** using services **EC2, ECR, IAM** for reliable and scalable access.
- Implemented **CI/CD pipelines** to automate testing, deployment, and continuous integration for faster updates and reduced downtime.

AI Voice Assistant – Jarvis | *Python, Gemini API, SpeechRecognition, pyttsx3, ReactJS, Tailwind CSS*

- Built an AI-powered voice assistant with context-aware dialogue using Gemini API, voice input via **SpeechRecognition** and **TTS** output with **pyttsx3**
- Integrated real-time features like **web search**, music playback and news updates for multi-command task automation.
- Developed a responsive frontend using **ReactJS** and **Tailwind CSS** enhancing user interaction and visual experience.

AI WhatsApp Chatbot- Lunexa | *Python, Gemini API, PyAutoGUI, Pyperclip*

- AI-powered **WhatsApp chatbot** that enables automated, human-like and context-aware conversations.
- Developed a conversational chatbot for WhatsApp Web using **Gemini API**, **PyAutoGUI** and **Pyperclip** to simulate natural interactions such as typing effects and loop-prevention.
- Enhanced user engagement by 95% through real-time, human-like and context-aware conversations

Education

K.K. Wagh Art's, Science and Commerce College Pimpalgaon (B). Nashik, Maharashtra
M.Sc. (Computer Science) | CGPA: 8.35 *August 2023 – June 2025*

K.K. Wagh Art's, Science and Commerce College Pimpalgaon (B). Nashik, Maharashtra
B.Sc. (Computer Science) | CGPA: 8.14 *August 2020 – July 2023*

Technical Skills

Languages : Python
AI/ML & LLMs : Generative AI, RAG, NLP, AI Agents, LangChain, OpenAI, LLaMA, Claude, Hugging Face, Gemini
Framework & Libraries: FastAPI, Flask, ReactJS, NumPy, Pandas, REST APIs
Databases : MySQL, PostgreSQL, Vector DBs (Pinecone, FAISS)
Cloud & DevOps: AWS (EC2, ECR, IAM), Docker, CI/CD Pipelines
Tools & Version Control: Git, GitHub, GitHub Actions